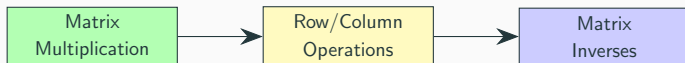


Row and Column Operations

Contravariant vs Covariant Logic

Learning Objectives



Key Questions:

- What are **row operations** and **column operations**?
- Why do rows have **contravariant** logic and columns have **covariant** logic?
- How do **simultaneous operations** preserve equations $AB = C$?
- How to use row/column operations to find **matrix inverses**?

Review: Three Views of Matrix Multiplication

For $C = AB$, we have three interpretations:

1. **Column view:** Each column of C is a linear combination of columns of A

$$c_j = Ab_j$$

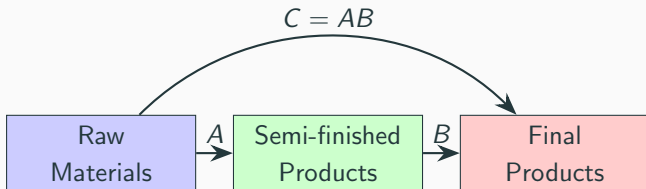
2. **Row view:** Each row of C is a linear combination of rows of B

$$c_i^T = a_i^T B$$

3. **Sum-of-rank-one:**

$$C = AB = \sum_{k=1}^n a_k b_k^T$$

Review: The Ingredient Table Interpretation



Matrix $C = AB$ represents the **composition** of two recipe tables:

- A : Semi-finished $\leftarrow$ Raw materials
- B : Final products $\leftarrow$ Semi-finished
- $C = AB$: Final products $\leftarrow$ Raw materials

Review: Coffee Shop Example

Matrix A (Semi-finished $\leftarrow$ Raw materials):

	 Milk	 Coffee	 Tea
 Leaf	0	0	2
 Lemon	0	0	1
 Bean	0	2	0
 Cow	1	0	0

$$A = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Interpretation:

- To make 1 cup of  Milk: need 1  Cow
- To make 1 cup of  Coffee: need 2  Beans
- To make 1 cup of  Tea: need 2  Leaves + 1  Lemon

Review: Coffee Shop Example (continued)

Matrix B (Final products $\leftarrow$ Semi-finished):

		Set 1		Set 2
$B =$				
 Milk		2	1	
 Coffee		0	2	
 Tea		1	1	

$$= \begin{pmatrix} 2 & 1 \\ 0 & 2 \\ 1 & 1 \end{pmatrix}$$

Interpretation:

-  Set 1: 2  Milks + 0  Coffees + 1  Tea
-  Set 2: 1  Milk + 2  Coffees + 1  Tea

Review: Coffee Shop Example (result)

Matrix $C = AB$ (Final products $\leftarrow$ Raw materials):

$$C = AB = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 0 & 4 \\ 2 & 1 \end{pmatrix}$$

	 Set 1	 Set 2
$C =$  Leaf	2	2
 Lemon	1	1
 Bean	0	4
 Cow	2	1

Interpretation:  Set 1 needs 2  , 1  , 0  , 2 

Where Each Concept Appears

Concept	Matrix A	Matrix B	Matrix C
Raw materials 	rows	—	rows
Semi-finished 	columns	rows	—
Final products 	—	columns	columns

The Fundamental Principle

When we **redefine** one of these three concepts (change the unit, repackage, bundle), all matrices that reference this concept must update their representation. But the underlying production relationships remain unchanged.

Row Operations: Contravariant Logic

Row Operations: Redefining Raw Materials

Core Idea: Row operations correspond to **redefining raw materials**.

When we redefine a raw material (change its unit or bundle it with others), we must update:

- The **rows of matrix A** (since raw materials appear in rows of A)
- The **rows of matrix C** (since raw materials appear in rows of C)
- Matrix B is **not affected** (semi-finished products don't change)

The Subtle Point

Row operations have **contravariant (inverse) logic** because rows and materials have a **multiplication relationship**:

$$(\text{demand coefficient}) \times (\text{material}) = (\text{total amount needed})$$

Why Contravariant Logic?

The key equation:

$$\text{demand} \times \text{material} = \text{constant}$$

Example: $4 \cdot x_2 = 2 \cdot (2x_2)$

- If material doubles: $x_2 \rightarrow 2x_2$
- Then demand must halve: $4 \rightarrow 2$
- Because: $2 \cdot (2x_2) = 4 \cdot x_2$ (same total)

Contravariant = Inverse Direction

- Material $\times k \Rightarrow$ Demand $\times \frac{1}{k}$
- Material increases $\Rightarrow$ Demand decreases

Type 1: Row Multiplication

Definition (Row Multiplication)

Replace row i with $k \cdot$ row i (where $k \neq 0$).

Notation: $r_i \rightarrow kr_i$

Interpretation: Redefine material $x_i \rightarrow \frac{1}{k}x_i$ (change unit size)

Contravariant Logic

- Material $\times \frac{1}{k} \Rightarrow$ Row $\times k$
- New unit is smaller $\Rightarrow$ Need more units

Example 1.1: Bean $\rightarrow$ Double Bean

Original table:

	 Set 1	 Set 2
 Leaf	2	2
 Lemon	1	1
 Bean	0	4
 Cow	2	1

Material redefinition: $x_3 \rightarrow 2x_3$ (one "double bean" = two old beans)

The new unit is **larger**, so we need **fewer** of them.

Example 1.1: Calculation

Original equation for  Set 2:

$$2 \cdot \text{🌿} + 1 \cdot \text{🍋} + 4 \cdot \text{🥵} + 1 \cdot \text{🐮}$$

With new material where (2 ) is the new unit:

$$2 \cdot \text{🌿} + 1 \cdot \text{🍋} + r_3 \cdot (2 \text{ 🥵}) + 1 \cdot \text{🐮}$$

For this to equal the original total:

$$r_3 \cdot (2 \text{ 🥵}) = 4 \cdot \text{🥵}$$

$$r_3 = 2$$

So we need 2 units of "double bean" (= 4 old beans). The demand coefficient decreased from 4 to 2.

Example 1.1: Result

Updated table (row 3 **divided by 2**):

	 Set 1	 Set 2
 Leaf	2	2
 Lemon	1	1
 Double bean	0	2
 Cow	2	1

Matrix notation: Perform $r_3 \rightarrow \frac{1}{2}r_3$:

$$\begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 0 & 4 \\ 2 & 1 \end{pmatrix} \xrightarrow{r_3 \rightarrow \frac{1}{2}r_3} \begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 0 & 2 \\ 2 & 1 \end{pmatrix}$$

Key

Material $\times 2 \rightarrow$ Row $\times \frac{1}{2}$ (contravariant!)

Type 2: Row Switching

Definition (Row Switching)

Swap row i and row j .

Notation: $r_i \leftrightarrow r_j$

Interpretation: Reordering materials in the table.

This doesn't change any production relationships, just the order.

Example:

$$\begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 0 & 4 \\ 2 & 1 \end{pmatrix} \xrightarrow{r_1 \leftrightarrow r_2} \begin{pmatrix} 1 & 1 \\ 2 & 2 \\ 0 & 4 \\ 2 & 1 \end{pmatrix}$$

Type 3: Row Addition

Definition (Row Addition)

Replace row i with row $i + k \times$ row j .

Notation: $r_i \rightarrow r_i + kr_j$

Interpretation: Redefine material $x_j \rightarrow x_j - kx_i$ (bundle material j with material i)

Contravariant Logic

- Material $x_j \rightarrow x_j - kx_i$ (includes kx_i in each unit)
- Row $r_i \rightarrow r_i + kr_j$ (need k more units of i)
- **Opposite signs!**

Example 1.3: Lemon $\rightarrow$ Lemon + 2 Leaf

Original table:

	 Set 1	 Set 2
 Leaf	2	2
 Lemon	1	1
 Bean	0	4
 Cow	2	1

Material redefinition: $x_2 \rightarrow x_2 + 2x_1$

(each lemon now comes bundled with 2 free tea leaves)

Example 1.3: Calculation

Original equation for  Set 1:

$$2 \cdot \text{🌿} + 1 \cdot \text{🍊} + 0 \cdot \text{🍫} + 2 \cdot \text{🐮}$$

With new bundled material ($\text{🍊} + 2 \text{🌿}$):

$$\begin{aligned} r_1 \cdot \text{🌿} + 1 \cdot (\text{🍊} + 2 \text{🌿}) + 0 \cdot \text{🍫} + 2 \cdot \text{🐮} \\ = r_1 \cdot \text{🌿} + 1 \cdot \text{🍊} + 2 \cdot \text{🌿} + 0 \cdot \text{🍫} + 2 \cdot \text{🐮} \\ = (r_1 + 2) \cdot \text{🌿} + 1 \cdot \text{🍊} + 0 \cdot \text{🍫} + 2 \cdot \text{🐮} \end{aligned}$$

For this to equal the original: $r_1 + 2 = 2$, so $r_1 = 0$.

Example 1.3: Result

Updated table (leaf row $r_1 \rightarrow r_1 - 2r_2$, **subtract not add**):

	 Set 1	 Set 2
 Leaf	0	0
 + 2 	1	1
 Bean	0	4
 Cow	2	1

Matrix notation: Perform $r_1 \rightarrow r_1 - 2r_2$:

$$\begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 0 & 4 \\ 2 & 1 \end{pmatrix} \xrightarrow{r_1 \rightarrow r_1 - 2r_2} \begin{pmatrix} 0 & 0 \\ 1 & 1 \\ 0 & 4 \\ 2 & 1 \end{pmatrix}$$

Key

Lemon becomes  + 2  \ Leaf demand **subtracts** 2 $\times$ lemon

Summary: Contravariant Logic for Rows

Why contravariant?

Because: $r_i x_i + r_j x_j = \text{constant}$

If one side changes, the other must change in opposite direction.

Material Change	Row Operation
$x_i \rightarrow kx_i$	$r_i \rightarrow \frac{1}{k} r_i$ (reciprocal)
$x_i \rightarrow x_i + kx_j$	$r_j \rightarrow r_j - kr_i$ (opposite sign!)

Remember

Rows work **backwards** from the material change!

Column Operations: Covariant Logic

Column Operations: Redefining Products

Core Idea: Column operations correspond to **redefining final products**.

When we redefine a final product (change its portion size or bundle it), we must update:

- The **columns of matrix B** (since products appear in columns of B)
- The **columns of matrix C** (since products appear in columns of C)
- Matrix A is **not affected** (raw materials don't change)

The Crucial Difference

Column operations have **covariant (direct) logic** because columns and products have an **equation relationship**:

$$(\text{product}) = (\text{recipe})$$

They appear on opposite sides of the equals sign!

Why Covariant Logic?

The key equation:

$$\text{product} = \text{recipe}$$

Example: $y_1 = c_1$ implies $2y_1 = 2c_1$

- If product doubles: $y_1 \rightarrow 2y_1$
- Then recipe must double: $c_1 \rightarrow 2c_1$
- Because: $2y_1 = 2c_1$ (same relationship)

Covariant = Same Direction

- Product $\times k \Rightarrow$ Recipe $\times k$
- Product increases $\Rightarrow$ Recipe increases

Type 1: Column Multiplication

Definition (Column Multiplication)

Replace column j with $k \cdot$ column j (where $k \neq 0$).

Notation: $c_j \rightarrow kc_j$

Interpretation: Redefine product $y_j \rightarrow ky_j$ (make k times the portion)

Covariant Logic

- Product $\times k \Rightarrow$ Column $\times k$
- Make more product $\Rightarrow$ Need more ingredients

Example 1.4: Set 1 $\rightarrow$ Double Set 1

Original table:

	 Set 1	 Set 2
 Leaf	2	2
 Lemon	1	1
 Bean	0	4
 Cow	2	1

Product redefinition: $y_1 \rightarrow 2y_1$ (make double portion of Set 1)

What happens to recipe? Original equation:

$$y_1 = 2 \cdot \text{Leaf} + 1 \cdot \text{Lemon} + 0 \cdot \text{Bean} + 2 \cdot \text{Cow}$$

With new product $2y_1$:

$$2y_1 = 2 \times (2 \cdot \text{Leaf} + 1 \cdot \text{Lemon} + 0 \cdot \text{Bean} + 2 \cdot \text{Cow})$$

Example 1.4: Result

Updated table (column 1 **multiplied by 2**):

	 	Double Set 1	 Set 2
 Leaf		4	2
 Lemon		2	1
 Bean		0	4
 Cow		4	1

Matrix notation: Perform $c_1 \rightarrow 2c_1$:

$$\begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 0 & 4 \\ 2 & 1 \end{pmatrix} \xrightarrow{c_1 \rightarrow 2c_1} \begin{pmatrix} 4 & 2 \\ 2 & 1 \\ 0 & 4 \\ 4 & 1 \end{pmatrix}$$

Key

Product $\times 2 \rightarrow$ Column $\times 2$ (**same direction!**)

Type 2: Column Switching

Definition (Column Switching)

Swap column i and column j .

Notation: $c_i \leftrightarrow c_j$

Interpretation: Reordering products in the table.

Example:

$$\begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 0 & 4 \\ 2 & 1 \end{pmatrix} \xrightarrow{c_1 \leftrightarrow c_2} \begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 4 & 0 \\ 1 & 2 \end{pmatrix}$$

Type 3: Column Addition

Definition (Column Addition)

Replace column j with column $j + k \times$ column i .

Notation: $c_j \rightarrow c_j + kc_i$

Interpretation: Redefine product $y_j \rightarrow y_j + ky_i$ (bundle product j with product i)

Covariant Logic

- Product $y_j \rightarrow y_j + ky_i$ (includes k units of product i)
- Column $c_j \rightarrow c_j + kc_i$ (add k times recipe of i)
- **Same signs!**

Example 1.6: Set 2 $\rightarrow$ Set 2 + Set 1

Original table:

	 Set 1	 Set 2
 Leaf	2	2
 Lemon	1	1
 Bean	0	4
 Cow	2	1

Product redefinition: $y_2 \rightarrow y_2 + y_1$ (bundle Set 2 with Set 1)

Original equations:

$$y_1 = 2 \text{  } + 1 \text{  } + 0 \text{  } + 2 \text{  }$$

$$y_2 = 2 \text{  } + 1 \text{  } + 4 \text{  } + 1 \text{  }$$

Example 1.6: Calculation

With new bundled product ($y_2 + y_1$):

$$\begin{aligned}(y_2 + y_1) &= (\text{recipe for } y_2) + (\text{recipe for } y_1) \\ &= (2 + 2) \text{ 🌿 } + (1 + 1) \text{ 🍋 } \\ &\quad + (4 + 0) \text{ ☕ } + (1 + 2) \text{ 🐮 } \\ &= 4 \text{ 🌿 } + 2 \text{ 🍋 } + 4 \text{ ☕ } + 3 \text{ 🐮 }\end{aligned}$$

Covariant Logic

Making the bundle requires ingredients for **both** products added together!

Example 1.6: Result

Updated table (column 2: $c_2 \rightarrow c_2 + c_1$, **add not subtract**):

	 Set 1	 Combo
 Leaf	2	4
 Lemon	1	2
 Bean	0	4
 Cow	2	3

Matrix notation: Perform $c_2 \rightarrow c_2 + c_1$:

$$\begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 0 & 4 \\ 2 & 1 \end{pmatrix} \xrightarrow{c_2 \rightarrow c_2 + c_1} \begin{pmatrix} 2 & 4 \\ 1 & 2 \\ 0 & 4 \\ 2 & 3 \end{pmatrix}$$

Key

Summary: Covariant Logic for Columns

Why covariant?

Because: $y_j = c_j$ (product equals its recipe)

If one side changes, the other must change in the same direction.

Product Change	Column Operation
$y_j \rightarrow ky_j$	$c_j \rightarrow kc_j$ (same scaling)
$y_j \rightarrow y_j + ky_i$	$c_j \rightarrow c_j + kc_i$ (same sign!)

Remember

Columns work **forwards** with the product change!

THE KEY DISTINCTION

	Rows (Materials)	Columns (Products)
Relationship	demand $\times$ material	product = recipe
Logic	Contravariant (inverse)	Covariant (direct)
Scaling	material $\times k$ $\Rightarrow$ row $\times \frac{1}{k}$	product $\times k$ $\Rightarrow$ column $\times k$
Addition	$x_i \rightarrow x_i + kx_j$ $\Rightarrow r_j \rightarrow r_j - kr_i$	$y_j \rightarrow y_j + ky_i$ $\Rightarrow c_j \rightarrow c_j + kc_i$

Why We Don't Update Semi-finished Products

Semi-finished products ( ,  , ) appear in:

- **Columns of A** (as outputs) $\rightarrow$ would need column operations (covariant)
- **Rows of B** (as inputs) $\rightarrow$ would need row operations (contravariant)

The Problem

Updating semi-finished products would require:

- **Two different types of operations**
- With **opposite logic** (contravariant vs covariant)
- On **two different matrices** (both A and B)

This is significantly more complex than row-only or column-only operations.

Solution: We avoid this complication and focus on raw materials (rows)

Simultaneous Operations

Preserve Equations

Change of Perspective

What we've learned so far:

- **Why** rows are contravariant (multiplication relationship)
- **Why** columns are covariant (equation relationship)
- **How** to update tables when redefining materials/products

From This Point Forward

We **forget about materials/products** and focus only on:

- **What:** Matrices A , B , C
- **How:** Simultaneous operations keep equations valid

This simplifies our work: we only track **matrix changes**, not material redefinitions.

Proposition 1.1: Row Operation Invariance

Proposition (Row Operation Invariance)

In the equation $C = AB$, if we perform the **same row operation** on both A and C simultaneously, the equality remains valid.

Why? Raw materials appear in the **rows** of both A and C . When we redefine raw materials, both must update. Matrix B doesn't reference raw materials, so it doesn't change.

But from now on, we don't need to think about "why" — we just use the fact that **simultaneous row operations on A and C preserve the equation**.

Example 1.7: Row Operation Preserves Equation

Consider:

$$\underbrace{\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}}_A \underbrace{\begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}}_B = \underbrace{\begin{pmatrix} 2 & 2 \\ 2 & 3 \end{pmatrix}}_C$$

Perform $r_1 \rightarrow 2r_1$ on **both A and C**:

$$\underbrace{\begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}}_{A'} \underbrace{\begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}}_B = \underbrace{\begin{pmatrix} 4 & 4 \\ 2 & 3 \end{pmatrix}}_{C'}$$

Still valid! ✓

Note: B is unchanged because it doesn't have raw materials in its rows.

Proposition 1.2: Column Operation Invariance

Proposition (Column Operation Invariance)

In the equation $C = AB$, if we perform the **same column operation** on both B and C simultaneously, the equality remains valid.

Why? Final products appear in the **columns** of both B and C . When we redefine final products, both must update. Matrix A doesn't reference final products, so it doesn't change.

But from now on, we don't need to think about products — we just use the fact that **simultaneous column operations on B and C preserve the equation**.

Example 1.8: Column Operation Preserves Equation

Consider:

$$\underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}}_A \underbrace{\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}}_B = \underbrace{\begin{pmatrix} 2 & 1 \\ 2 & 6 \end{pmatrix}}_C$$

Perform $c_1 \rightarrow 3c_1$ on **both B and C**:

$$\underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}}_A \underbrace{\begin{pmatrix} 6 & 1 \\ 3 & 3 \end{pmatrix}}_{B'} = \underbrace{\begin{pmatrix} 6 & 1 \\ 6 & 6 \end{pmatrix}}_{C'}$$

Still valid! ✓

Note: A is unchanged because it doesn't have final products in its columns.

Why These Propositions Matter

Foundation for Matrix Equations

Propositions 1.1 and 1.2 provide the theoretical basis for:

- Solving matrix equations $AB = C$
- Finding matrix inverses A^{-1}
- Understanding how operations affect equations

Key insight: We can **manipulate** equations $C = AB$ using row/column operations, just like we manipulate algebraic equations, as long as we:

- Apply row operations to **both A and C**
- Apply column operations to **both B and C**

Matrix Inverses

Definition: Matrix Inverse

Definition (Matrix Inverse)

An $n \times n$ matrix A is called **invertible** if there exists an $n \times n$ matrix B such that

$$AB = BA = I_n$$

where I_n is the $n \times n$ identity matrix. We call B the **inverse** of A and denote it A^{-1} .

Interpretation: If A is the recipe "raw materials $\rightarrow$ products", then A^{-1} is the recipe "products $\rightarrow$ raw materials".

Example 1.9: Coffee Shop Inverse

Consider:

$$A = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} = \begin{array}{c|cc} & \img alt="coffee cup" data-bbox="540 185 585 240"/> \text{Coffee} & \img alt="milk glass" data-bbox="700 185 735 245"/> \text{Milk} \\ \hline \img alt="bean" data-bbox="395 260 440 315"/> \text{Bean} & 0 & 2 \\ \img alt="cow" data-bbox="395 335 445 400"/> \text{Cow} & 1 & 0 \end{array}$$

Then:

$$A^{-1} = \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & 0 \end{pmatrix} = \begin{array}{c|cc} & \img alt="bean" data-bbox="560 515 605 575"/> \text{Bean} & \img alt="cow" data-bbox="705 510 755 575"/> \text{Cow} \\ \hline \img alt="coffee cup" data-bbox="400 590 445 645"/> \text{Coffee} & 0 & 1 \\ \img alt="milk glass" data-bbox="420 665 450 730"/> \text{Milk} & \frac{1}{2} & 0 \end{array}$$

Verification:

$$AA^{-1} = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

Proposition 1.3: Inverse is Unique

Proposition (Uniqueness of Inverse)

If A has an inverse, it is unique.

Proof: Suppose B and C both satisfy the definition. Then:

$$B = BI = B(AC) = (BA)C = IC = C$$

Therefore $B = C$.



Proposition 1.4: Left Inverse = Right Inverse

Proposition (One-sided Inverse Suffices)

For a square matrix A , if $AB = I$, then $B = A^{-1}$ (and automatically $BA = I$).

Proof: Suppose $AB = I$. We need to show $BA = I$.

Since A is square and $AB = I$, we know A has a left inverse. Let $CA = I$ for some matrix C .

Then:

$$B = IB = (CA)B = C(AB) = CI = C$$

Therefore $BA = CA = I$. So $AB = BA = I$, which means $B = A^{-1}$. $\square$

Practical Implication of Proposition 1.4

Important!

To verify that B is the inverse of square matrix A , you only need to check **one direction**!

If $AB = I$, then automatically $BA = I$.

This saves computation time when verifying inverses.

Example: To check if $\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$ is the inverse of $\begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$:

Just compute one product (either AB or BA). If it equals I , done!

Proposition 1.5: Product of Invertible Matrices

Proposition (Inverse of Product)

If A and B are invertible, then AB is invertible and

$$(AB)^{-1} = B^{-1}A^{-1}$$

Proof:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

By Proposition 1.4, this proves $(AB)^{-1} = B^{-1}A^{-1}$. □

Order Reversal

$(AB)^{-1} = B^{-1}A^{-1}$, **not** $A^{-1}B^{-1}$.

Intuition: Socks then shoes $\rightarrow$ undo by removing shoes first, then socks.

Why Introduce Inverses?

Why Do We Need Matrix Inverses?

Question: Why introduce the concept of inverse matrices?

Answer: Inverses provide **notational convenience** for tracking invariants when solving matrix equations.

The Key Insight

Writing expressions like $B = A^{-1}C$ or $B = CA^{-1}$ makes explicit that:

- " **B doesn't change**" when we perform operations
- We can **simultaneously transform** the " A^{-1} " part and the " C " part
- The transformation preserves the equation

The next two theorems are the **most important** results for practical computation!

Theorem 1.6: Row Operations on $A^{-1}C$

Proposition (Row Operations Preserve $A^{-1}C$)

In the expression $B = A^{-1}C$, if we perform the **same row operation** on both "A" and "C" simultaneously, we get:

$$B = (A')^{-1}C'$$

where A' , C' are the matrices after the row operation, and **B remains unchanged.**

Why? From $AB = C$, applying row operations to both A and C preserves the equation (Prop 1.1):

$$A'B = C'$$

Therefore: $B = (A')^{-1}C'$

Key Application

This theorem allows us to solve $AB = C$ for B by performing row operations!

Example 1.10: Using $A^{-1}C$ Row Operations (Setup)

Problem: Solve $AB = C$ for B where

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 5 & 8 \\ 3 & 5 \end{pmatrix}$$

Method: Write $B = A^{-1}C$ and perform row operations to transform " A " into I :

$$B = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 5 & 8 \\ 3 & 5 \end{pmatrix}$$

Goal: Transform " A " to " I " using row operations.

Example 1.10: Step 1-2

Step 1: $r_1 \leftrightarrow r_2$

$$B = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 3 & 5 \\ 5 & 8 \end{pmatrix}$$

Step 2: $r_2 \rightarrow r_2 - 2r_1$

$$B = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}^{-1} \begin{pmatrix} 3 & 5 \\ -1 & -2 \end{pmatrix}$$

Example 1.10: Step 3-4

Step 3: $r_2 \rightarrow (-1) \cdot r_2$

$$B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$$

Step 4: $r_1 \rightarrow r_1 - r_2$

$$B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} = I^{-1} \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$$

Since $I^{-1} = I$:

$$B = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$$

Example 1.10: Verification

Check: Does $AB = C$?

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 4+1 & 6+2 \\ 2+1 & 3+2 \end{pmatrix} = \begin{pmatrix} 5 & 8 \\ 3 & 5 \end{pmatrix}$$

Yes! ✓

Key Observation

We transformed "A" to "I" using row operations, and automatically obtained B!

Theorem 1.7: Column Operations on CB^{-1}

Proposition (Column Operations Preserve CB^{-1})

In the expression $A = CB^{-1}$, if we perform the **same column operation** on both "C" and "B" simultaneously, we get:

$$A = C'(B')^{-1}$$

where C' , B' are the matrices after the column operation, and **A remains unchanged**.

Why? From $AB = C$, applying column operations to both B and C preserves the equation (Prop 1.2):

$$AB' = C'$$

Therefore: $A = C'(B')^{-1}$

Key Application

This theorem allows us to solve $AB = C$ for A by performing column operations!

Example 1.11: Using CB^{-1} Column Operations (Setup)

Problem: Solve $AB = C$ for A where

$$B = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 5 & 3 \\ 8 & 5 \end{pmatrix}$$

Method: Write $A = CB^{-1}$ and perform column operations to transform " B " into I :

$$A = \begin{pmatrix} 5 & 3 \\ 8 & 5 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}^{-1}$$

Goal: Transform " B " to " I " using column operations.

Example 1.11: Step 1-2

Step 1: $c_1 \leftrightarrow c_2$

$$A = \begin{pmatrix} 3 & 5 \\ 5 & 8 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}^{-1}$$

Step 2: $c_2 \rightarrow c_2 - 2c_1$

$$A = \begin{pmatrix} 3 & -1 \\ 5 & -2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}^{-1}$$

Example 1.11: Step 3-4

Step 3: $c_2 \rightarrow (-1) \cdot c_2$

$$A = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}^{-1}$$

Step 4: $c_1 \rightarrow c_1 - c_2$

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} I^{-1}$$

Wait, error! Let me recalculate...

Actually:

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$$

Summary: The Two Critical Theorems

Theorem 1.6: Row Operations on $A^{-1}C$

Simultaneous row operations on " A " and " C " in $B = A^{-1}C$ preserve B .

Use case: Solving $AB = C$ for B

Theorem 1.7: Column Operations on CB^{-1}

Simultaneous column operations on " C " and " B " in $A = CB^{-1}$ preserve A .

Use case: Solving $AB = C$ for A

Why These Matter

These two theorems are the **foundation** for:

- Finding matrix inverses
- Solving matrix equations
- Understanding matrix transformations

Finding Matrix Inverses

Finding A^{-1} : The Method

Goal: Find A^{-1} for a given matrix A .

Idea: Solving $AA^{-1} = I$ for A^{-1} is a special case of solving $AB = C$.

From $A^{-1} = A^{-1}I$, perform **row operations** to transform " A " into I :

$$A^{-1} = A^{-1}I \xrightarrow{\text{row ops}} I^{-1} \cdot (\text{result}) = \text{result}$$

Algorithm:

1. Write $A^{-1} = A^{-1}I$
2. Apply row operations to both " A " and " I " until " A " becomes I
3. The result on the right is A^{-1}

Example 1.12: Finding Inverse (Row Operations)

Find: A^{-1} where $A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$

Setup:

$$A^{-1} = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Step 1: $r_2 \rightarrow r_2 - r_1$

$$A^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$$

Example 1.12: Steps 2-5

Step 2: $r_1 \leftrightarrow r_2$

$$A^{-1} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}^{-1} \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix}$$

Step 3: $r_2 \rightarrow r_2 - 2r_1$

$$A^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}^{-1} \begin{pmatrix} -1 & 1 \\ 3 & -2 \end{pmatrix}$$

Step 4: $r_2 \rightarrow (-1) \cdot r_2$

$$A^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} -1 & 1 \\ -3 & 2 \end{pmatrix}$$

Step 5: $r_1 \rightarrow r_1 - r_2$

$$A^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$$

Example 1.13: Finding Inverse (Column Operations)

Alternative: Use column operations on $IA^{-1} = A^{-1}$

Setup:

$$A^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}^{-1}$$

Step 1: $c_1 \rightarrow c_1 - c_2$

$$A^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}^{-1}$$

Step 2: $c_2 \rightarrow c_2 - c_1$

$$A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}^{-1}$$

Example 1.13: Final Step

Step 3: $c_1 \rightarrow c_1 - c_2$

$$A^{-1} = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$$

Same answer! ✓

Observation

Both row operations and column operations give the same inverse.

When Does an Inverse Exist?

Not All Matrices Are Invertible

Example:

$$A = \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix}$$

Observation: Row 2 = 2 × Row 1 (redundant information)

Try to find inverse:

$$A^{-1} = \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Perform $r_2 \rightarrow r_2 - 2r_1$:

$$A^{-1} = \begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$$

Stuck!

Cannot transform "A" to I . This matrix has **no inverse**.

Invertibility Criterion (Preview)

Criterion

An $n \times n$ matrix is invertible if and only if it can be transformed to I using row operations.

Equivalently: **rank** = n .

Detailed discussion of rank and solvability conditions will be covered in **Chapter 2**.

For now:

- If row operations can reduce A to I : invertible
- If we get a zero row: not invertible

Visual Summary: Invertible vs Non-invertible

Invertible	Non-invertible
$\begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$	$\begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix}$
Can reduce to I	Gets zero row
Rows are independent	Rows are dependent
Full rank	Not full rank
Unique inverse A^{-1}	No inverse

Exercises

Exercise 1.1

Verify Proposition 1.4 for:

$$A = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$$

- (a) Check that $AB = I$
- (b) Verify that $BA = I$ also holds

Exercise 1.2

Consider the equation $AB = C$ where

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 4 & 7 \\ 3 & 5 \\ 2 & 3 \end{pmatrix}$$

Solve for B using row operations on $B = A^{-1}C$.

Exercise 1.3

Find the inverses using row operations:

$$(a) A = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$$

$$(b) A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

Exercise 1.4

Find the inverse of $A = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$ using **column operations**.

Verify it matches the result from row operations.

Exercise 1.5

Try to find the inverse of $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$.

What happens? Explain why this matrix is not invertible.

Exercise 1.6

Show that if A and B are invertible, then:

(a) $(A^{-1})^{-1} = A$

(b) $(A^T)^{-1} = (A^{-1})^T$

Exercise 1.7

Consider the coffee shop example with matrix

$$C = \begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 0 & 4 \\ 2 & 1 \end{pmatrix}$$

- (a) Perform row operation $r_1 \rightarrow r_1 - 2r_2$ and interpret the material redefinition.
- (b) Perform column operation $c_2 \rightarrow c_2 + c_1$ and interpret the product redefinition.

Summary: Key Takeaways

1. **Row operations: Contravariant** (inverse logic)
 - Material $\times k \rightarrow$ Row $\times \frac{1}{k}$ (reciprocal)
 - Material $x_i \rightarrow x_i + kx_j \rightarrow$ Row $r_j \rightarrow r_j - kr_i$ (opposite sign!)
2. **Column operations: Covariant** (direct logic)
 - Product $\times k \rightarrow$ Column $\times k$ (same)
 - Product $y_j \rightarrow y_j + ky_i \rightarrow$ Column $c_j \rightarrow c_j + kc_i$ (same sign!)
3. **Equation $C = AB$ is invariant** under:
 - Simultaneous row ops on A and C (Prop 1.1)
 - Simultaneous column ops on B and C (Prop 1.2)

Summary: Critical Theorems

4. **Left inverse = right inverse** for square matrices (Prop 1.4)
5. **Row operations on $A^{-1}C$ preserve B** (Theorem 1.6)
 - Use to solve $AB = C$ for B
6. **Column operations on CB^{-1} preserve A** (Theorem 1.7)
 - Use to solve $AB = C$ for A
7. **Finding A^{-1}** : use row ops on $A^{-1}I$ or column ops on IA^{-1} until " A " becomes I

Remember

Rows work backwards. Columns work forwards.